

Acquisition and Analysis of Neural Data

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Problem Set 3

Deadline: Tuesday 5th May, 2026 at 23:55

Please work in groups of ideally 2 students. Every group member must submit a copy of their group solutions and include the names of all group members in the files. Please specify how much time you needed to finish each part and a statement on whether you used AI tools (otherwise, 5 points will be subtracted).

Analytical Part: scan and upload your solution as a single PDF. Make sure all text is clearly readable.

Numerical Part: submit both a Python code file (or notebook) and a PDF or HTML file including figures and code. Use clear, descriptive variable names and comments.

Analytical Part - Spike-train auto-correlation and cross-correlation

The spike train auto-correlation function

$$Q_{\rho\rho}(\tau) = \frac{1}{T} \int_0^T dt \langle \rho(t)\rho(t+\tau) \rangle - \langle r \rangle^2 \quad (1)$$

is used to **detect patterns in spike trains** (e.g., oscillations) and to **classify cells**. Similarly, the spike train cross-correlation function

$$Q_{\rho\rho'}(\tau) = \frac{1}{T} \int_0^T dt \langle \rho(t)\rho'(t+\tau) \rangle - \langle r \rangle \langle r' \rangle \quad (2)$$

is used to **find relationships between a pair of spike trains**, e.g., **to assess whether two neurons fire synchronously** or **phase-locked**, which could be evidence for **synaptic connectivity**.

1. Auto-correlation histogram of a Poisson process (3 points)

Sketch the expected (i.e., trial averaged) unnormalized auto-correlation histogram N_m (defined in the lecture, see also Dayan and Abbott, Section 1.4, text after Equation 1.35) of a Poisson process with refractory **period $t_{\text{ref}} = 3$ ms** in the **interval $[-20, 20]$ ms**. Consider **bin size $\Delta t = 0.1$ ms**, **trial length $T = 1$ s**, and **firing rate $\langle r \rangle = 100$ spikes/s**, where $\langle r \rangle$ is the effective rate of the process after introducing refractoriness (see also numerical exercise 3). Repeat the sketch for $\langle r \rangle = 300$ spikes/s (high rate) and $\langle r \rangle = 10$ spikes/s (low rate).

Hint: Start by calculating the driving rate of the Poisson process corresponding to each given effective rate. The relation between both quantities affects what happens in each of the three cases.

Note: Although approximate, your sketches should include relevant values on the appropriate axes, such that they clearly show the answer to the following questions:

- (i) For what values of m is $N_m = 0$?
- (ii) For $m \neq 0$, what is the maximum value of N_m and when does it occur?
- (iii) What value does N_m approach for $t_{\text{ref}} \ll |m|\Delta t \ll T$?

2. Cross-correlation of oscillatory signals (4 points)

Calculate the cross-correlation function $Q_{r_1 r_2}(\tau)$ between two oscillatory firing rates

$$\begin{aligned} r_1(t) &= [1 + a_1 \sin(\omega t)] \bar{r}_1, \\ r_2(t) &= [1 + a_2 \sin(\omega t + \phi)] \bar{r}_2 \end{aligned}$$

where $\bar{r}_1, \bar{r}_2$ are constant firing rates, ω is an angular frequency, ϕ is a phase shift, and $0 < a_1 < 1$ and $0 < a_2 < 1$ are dimensionless constants. Approximate Q for the case in which the integration time T is much longer than the

length of one oscillation cycle, i.e., $T \gg 2\pi/\omega$. Hint: Use the definition in Equation 2 (without trial averaging) and remember the following trigonometric identity

$$\sin(A)\sin(B) = \frac{1}{2} [\cos(A-B) - \cos(A+B)].$$

3. Symmetry of the auto-correlation (3 points)

- (a) Prove the symmetry property of the spike-train auto-correlation function, i.e., $Q_{pp}(\tau) = Q_{pp}(-\tau)$.
Hint: Use variable substitution.
- (b) Briefly discuss the integration limits.

Numerical Part - Spike-train statistics

The Poisson process plays a central role in the analysis of spiking neurons and networks. In this exercise, you get familiar with some basic properties of the Poisson process. Feel free to recycle your previous code (at least if it gave you the correct results).

1. Coefficient of variation and Fano factor (4 points)

The coefficient of variation CV and the Fano factor F are measures that characterize the variability (or regularity) of a spike train. Both measures quantify the dispersion of the data in relation to its mean.

The coefficient of variation is defined as:

$$CV = \frac{\sigma_\tau}{\langle \tau \rangle}$$

where σ_τ and $\langle \tau \rangle$ are the standard deviation and the mean of the **inter-spike intervals τ (ISIs)**. The CV can be computed within a trial, or pooling ISIs across trials.

The Fano factor is defined as:

$$F = \frac{\sigma_n^2}{\langle n \rangle}$$

where σ_n^2 and $\langle n \rangle$ are the variance and the mean of **the spike counts n in time windows of length T_W** . If the counting window spans the entire trial ($T_W = T$), spike counts n are pooled across trials and F is a single number. By contrast, if $T_W < T$, each trial is subdivided in $N_W = \lfloor T/T_W \rfloor$ non-overlapping windows of length T_W . In this case, the Fano factor can be either *trial resolved* or *time resolved*. In the first case, spike counts are pooled within trials and across windows (trial resolved, one F value per trial). In the latter case, spike counts are pooled within windows and across trials (time resolved, one F value per time window).

Download `SpikeTimes.dat` from the Moodle webpage and load it with the numpy function `loadtext`. The file contains a matrix of spike times in milliseconds. The dimensions are $[m \times n]$, where $m=100$ is the maximum spike-time index and $n=100$ is the number of trials. Note: each trial has a different number of spikes, the remaining entries are filled with `nan`. **The trial length is $T = 5.5$ s, the resolution is 0.1 ms.** It is convenient to convert the data in units of seconds.

- (a) Make a raster plot of the data. Use the pyplot function `eventplot`. You may need to set the axes limits manually for correct visualization.
- (b) Plot a histogram of **the ISIs pooled across trials. Hint: use the numpy function `diff`**. How are the ISIs distributed?
- (c) Compute the coefficient of variation CV from the pooled ISIs.
- (d) Compute and plot both the trial-resolved and the time-resolved Fano factor F in non-overlapping windows of lengths $T_W \in [0.3, 0.5, 1]$ s. Interpret the results. How does the window size affect the Fano factor in this dataset?

2. Homogeneous and inhomogeneous Poisson process (4 points)

Download and load the file **`PoissonSpikeTrains.mat`** using the function **`loadmat` in `scipy.io`**. The arrays **`SpikeTimes_hom`** and **`SpikeTimes_inh`** contain spike times in **milliseconds** for an homogeneous and an inhomogeneous Poisson process, respectively. The Python program `generatePoissonTrains.py` explains how these spike times were generated. For both the homogeneous and the inhomogeneous process:

- (a) Construct and plot the ISIs histograms. How are they different?
- (b) Compute the CV and the Fano factor. For the fano Factor use **non-overlapping windows of length $T_W = 100$ ms** (only one trial).
- (c) Compute and plot the spike-train auto-correlation histogram by discretizing the auto-correlation function

$$Q_{pp}(\tau) = \frac{1}{T} \int_0^T dt \langle p(t)p(t+\tau) \rangle - \langle r \rangle^2.$$

Use a bin size **$\Delta t = 0.1$ ms and multiply the result by Δt** such that the auto-correlation in zero equals the average firing rate $\langle r \rangle$. Use the **Numpy function correlate**. Plot the auto-correlation both **with and without** the central peak. Also try different x-axis limits to see if you can observe any interesting structure. If you want, you can additionally implement step-by-step the algorithm outlined in Dayan and Abbott (Section 1.4, text after Equation 1.35). With the correct scaling, the two procedures should lead to the same result.

- (d) Why do the CV and the Fano factor deviate from the theoretical value of 1?

3. Poisson process with refractory period (2 points)

Load the variables **SpikeTimes_ref** and **rates_ref** from the data file PoissonSpikeTrains.mat. The array SpikeTimes_ref contains the **spike times in milliseconds** generated from **six** homogeneous Poisson processes with refractoriness. Each process has a different **driving rate, i.e.**, the rate before refractoriness is introduced (more details in generatePoisson-Trains.py). The driving rates are given in the variable rates_ref. **The refractory period is 5 ms.**

- (a) Estimate the effective firing rate r_{eff} for each driving rate r_{drive} . Plot r_{eff} against r_{drive} . Which level does r_{eff} approach for large r_{drive} ?
- (b) Construct and plot the ISI histograms for each driving rate.
- (c) Compute the CV and the Fano factor for each driving rate and plot them against r_{drive} . Why do they decrease with increasing rates?
- (d) Calculate and plot the spike-train autocorrelation function for each driving rate. Use a bin size $\Delta t = 1$ ms. Why are there multiple peaks for large driving rates?

4. Extra: Gamma Process (3 extra points)

A Gamma process of order k (with a constant rate) can be generated from a homogeneous Poisson process by retaining only each k^{th} spike.

- (a) Generate Gamma processes of order 2, 3, 5, and 10, each with a rate of $r = 100$ spikes/s (the rate of the underlying homogeneous Poisson process needs to be adjusted accordingly). For each Gamma process, generate 1000 spikes.
- (b) Compute ISI histograms and CVs. Compare your results to the theoretical distribution

$$p_{\text{ISI}}(\tau) = \frac{r(r\tau)^{k-1}e^{-r\tau}}{(k-1)!}$$

where τ is the inter-spike interval and k is the order of the Gamma process.

- (c) Compile your results into a figure where the left column shows example spike trains of 1 s length of each Gamma process, and the right column shows the corresponding ISI distributions, with the CV values next to them.